Feng Ruan

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Academic and Research Employment

Assistant Professor of Statistics and Data Science. Northwestern University 2022-present.

Education

Postdoctoral Researcher. University of California, Berkeley, 2022. Advisor: Michael I . Jordan.

Ph.D. Department of Statistics, Stanford, 2019. Advisor: John C. Duchi.

B.S. (with distinction) Department of Mathematics, Peking University, 2014.

B.A. (double degree) China Center for Economic Research, Peking University, 2014.

Publications

Journal Publications

1. Y. Li and F. Ruan. A variational analysis of kernel learning with learnable linear transformations. *Foundation of Computational Mathematics, to appear*, 2026
2. F. Ruan, K. Liu, and M. Jordan. A compositional kernel model for feature learning. *Applied and Computational Harmonic Analysis, to appear*, 2026.
3. F. Ruan. On the uniform convergence of subdifferentials in stochastic optimization and learning. *Mathematics of Operations Research*, 2025
4. X. Li, F. Ruan, H. Wang, Q. Long, and W. J. Su. A statistical framework of watermarks for large language models: Pivot, detection efficiency and optimal rules. *Annals of Statistics*, 2025
5. A. Montanari, F. Ruan, Y. Sohn, and J. Yan. The generalization error of max-margin linear classifiers: High-dimensional asymptotics in the overparametrized regime. *Annals of Statistics*, 2025
6. X. Li, F. Ruan, H. Wang, Q. Long, and W. J. Su. Robust detection of watermarks for large language models under human edits. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, page qkaf056, 2025
7. B. Kendall, C. Jin, K. Li, F. Ruan, X. Wang, and J.-P. Wang. Dnacycp2: improved estimation of intrinsic dna cyclizability through data augmentation. *Nucleic Acids Research*, 53(5):gkaf145, 2025
8. Y. Chen, F. Ruan, and J.-P. Wang. Nlsdeconv: an efficient cell-type deconvolution method for spatial transcriptomics data. *Bioinformatics*, 41(1):btae747, 2025
9. J. C. Duchi and F. Ruan. The right complexity measure in locally private estimation: It is not the fisher information. *Annals of Statistics*, 52(1):1–51, 2024
10. J. C. Duchi and F. Ruan. Asymptotic optimality in stochastic optimization. *Annals of Statistics*, 49(1):21–48, 2021
11. I. Lemhadri, F. Ruan, and R. Tibshirani. Lassonet: A neural network with feature sparsity. *Journal of Machine Learning Research*, 22(127):1–29, 2021

12. K. Khamaru, A. Pananjady, F. Ruan, M. J. Wainwright, and M. I. Jordan. Is temporal difference learning optimal? An instance-dependent analysis. *SIAM Journal on Mathematics of Data Science*, 2021
13. L. T. Liu, F. Ruan, H. Mania, and M. I. Jordan. Bandit learning in decentralized matching markets. *Journal of Machine Learning Research*, 2021
14. J. C. Duchi, K. Khosravi, and F. Ruan. Multiclass classification, information, divergence, and surrogate risk. *Annals of Statistics*, 46:3246–3275, 2018
15. J. C. Duchi and F. Ruan. Solving (most) of a set of quadratic equalities: Composite optimization for robust phase retrieval. *Information and Inference*, iay015, 2018
16. J. C. Duchi and F. Ruan. Stochastic methods for composite and weakly convex optimization problems. *SIAM Journal on Optimization*, 28:3229–3259, 2018
17. B. Kaneshiro, F. Ruan, C. W. Baker, and J. Berger. Characterizing listener engagement with popular songs using large-scale music discovery data. *Frontiers in Psychology*, 8:416, 2017
18. S. Osher, F. Ruan, J. Xiong, Y. Yao, and W. Yin. Sparse recovery via differential inclusions. *Applied and Computational Harmonic Analysis*, 41(2):436–469, 2016

Conference Publications

1. H. Asi, J. C. Duchi, S. Haque, Z. Li, and F. Ruan. Universal instance-optimal mechanisms for private statistical estimation. In *Proceedings of the Thirty Seventh Annual Conference on Computational Learning Theory (COLT)*, 2024
2. I. Lemhadri, F. Ruan, and R. Tibshirani. Lassonet: Neural networks with feature sparsity. In *International Conference on Artificial Intelligence and Statistics*, pages 10–18. PMLR, 2021
3. J. Duchi and F. Ruan. A constrained risk inequality for general losses. In *International Conference on Artificial Intelligence and Statistics*, pages 802–810. PMLR, 2021. (oral presentation)
4. J. C. Duchi, F. Ruan, and C. Yun. Minimax bounds on stochastic batched convex optimization. In *Proceedings of the Thirty First Annual Conference on Computational Learning Theory (COLT)*, 2018. (27.2% acceptance rate).
5. J. C. Duchi, K. Khosravi, and F. Ruan. Information measures, experiments, multi-category hypothesis tests, and surrogate losses. In *The 54th Allerton Conference on Communication, Control, and Computing*, 2016.

Preprints & Papers under Review

1. A. Montanari, F. Ruan, B. Saeed, and Y. Sohn. Universality of max-margin classifiers. *arXiv preprint arXiv:2310.00176 (Major Revision at Mathematical Statistics and Learning)*, 2023.
2. M. I. Jordan, K. Liu, and F. Ruan. On the self-penalization phenomenon in feature selection. *arXiv preprint arXiv:2110.05852 (Major Revision at Journal of Machine Learning Research)*, 2021
3. A. Montanari, F. Ruan, and J. Yan. Adapting to unknown noise distribution in matrix denoising. *arXiv preprint arXiv:1810.02954 (Major revision at Bernoulli)*, 2018.
4. Y. Chen, Y. Li, K. Liu, and F. Ruan. A theory of feature learning in kernel models. *arXiv preprint arXiv:2310.11736*, 2023.
5. K. Liu and F. Ruan. On the limitation of kernel dependence maximization for feature selection. *arXiv preprint arXiv:2406.06903*, 2024.

6. Y. Li and F. Ruan. Gradient flow in the kernel learning problem. *arXiv preprint arXiv:2506.08550*, 2025
7. K. Liu and F. Ruan. A self-penalizing objective function for scalable interaction detection. *arXiv preprint arXiv:2011.12215*, 2020.

Softwares

1. F. Ruan, J. Xiong and Y. Yao R Package *LIBRA*
   available at `https://cran.r-project.org/web/packages/Libra/index.html`
2. L. Abraham, I. Lemhadri, F. Ruan and R. Tibshirani Python Package *LassoNet*
   available at `https://lassonet.ml/`
3. K. Liu and F. Ruan Python Package *Metric Learning*
   available at `https://keli-feng.github.io/package/metric-learning.zip`
4. K. Liu, F. Ruan and R. Tibshirani R Package *FCLUSTER*.

Funding

National Science Foundation CAREER, DMS-2540678, *Beyond Multi-Index Models: Statistical and Algorithmic Foundations for Feature Learning in Compositional Architectures*, 2026–2031, $416,950.

Awards

National Science Foundation CAREER Award, 2025.

Departmental Teaching Assistant Award, Department of Statistics, Stanford University, 2017.

Stanford Graduate Fellowship (E.K. Potter Fellowship), Stanford University, 2014.

Outstanding Graduates in Prominent Talents in Applied Mathematics Training Program, Peking University, 2014.

Bronze Model in Chinese Mathematical Olympiad, 2010.

Silver Medal in Romanian Master of Mathematical Olympiad, 2009.

Teaching Experience

Statistics 430-2, Probability for Statistical Inference 1, Fall 2025. Northwestern University.

Statistics 350-0, Regression Analysis, Fall 2025. Northwestern University.

Statistics 320-2, Statistical Theory & Methods 2, Winter 2025. Northwestern University.

Statistics 430-2, Probability for Statistical Inference 2, Winter 2025. Northwestern University.

Statistics 350-0, Regression Analysis, Fall 2024. Northwestern University.

Statistics 320-2, Statistical Theory and Methods, Spring 2024. Northwestern University.

Statistics 350-0, Regression Analysis, Fall 2023. Northwestern University.

Statistics 430-1, Probability for Statistical Inference, Fall 2023. Northwestern University.

Statistics 320-2, Statistical Theory and Methods, Spring 2023. Northwestern University.

Statistics 461-0, Information Theory and Statistics, Spring 2023. Northwestern University.

Statistics 430-1, Probability for Statistical Inference, Fall 2022. Northwestern University.

Talks and Presentations

Conference

2026	IMS-APRM (Institute of Mathematical Statistics—Asia Pacific Rim Meeting)
2026	SIAM Conference on Optimization (OP26)
2025	INFORMS Annual Meeting
2025	International Conference on Continuous Optimization
2024	INFORMS Annual Meeting
2024	International Symposium on Mathematical Programming
2024	Conference on Information Science and Systems
2023	International Conference Stochastic Programming
2023	SIAM Conference on Optimization
2023	INFORMS Annual Meeting
2023	Midwest Machine Learning Symposium
2023	Joint Statistical Meetings (JSM)
2022	INFORMS Annual Meeting
2020	Artificial Intelligence and Statistics (AIStats) (online)
2020	Joint Statistical Meetings (JSM) (online)
2018	Conference on Learning Theory (COLT) (Stockholm, Sweden)
2017	IMS/ASA Spring Research Conference (Rutgers University, New Brunswick, NJ, USA)

Seminars

2025	Department of Mathematics, Analysis Seminar, Northwestern University
2025	Department of Mathematics, New Jersey Institute of Technology
2023	Chicago Booth School of Business, University of Chicago
2022	Department of Statistics and Data Science, Cornell University
2022	IEMS Department Seminar, Northwestern University
2022	ISYE Department Seminar, Georgia Institute of Technology
2022	Applied Mathematics Department Seminar, Brown University
2022	Math Department Seminar, Courant Institute of Mathematical Sciences
2022	Statistics Department Seminar, The Wharton School at the University of Pennsylvania
2022	Statistics Department Seminar, University of Minnesota Twin Cities
2022	Statistics Department Seminar, Northwestern University
2022	Statistics & Operations Research Seminar, Krannert School of Management at Purdue University
2022	Statistics Department Seminar, Northwestern University
2022	Statistics Department Seminar, Fuqua School of Business at Duke University
2021	Statistics Department Seminar, University of California, Berkeley
2021	Statistics Department Seminar, ETH Zurich
2021	Statistics Group Seminar, HongKong University of Science and Technology
2021	International Seminar on Selective Inference
2020	Information Theory Group Seminar (BLISS), University of California, Berkeley
2019	Information Theory Forum, Stanford University
2019	Biostatistics Department Seminar, University of California, Berkeley
2019	Data Science Institute Department Seminar, University of California, San Diego
2019	Applied Mathematics and Statistics Department Seminar, Johns Hopkins University
2019	ISYE Department Seminar, Georgia Institute of Technology
2019	EE Department Seminar, University of Michigan
2019	EE Department Seminar, Princeton University
2019	Statistics Department Seminar, University of Wisconsin-Madison
2019	Statistics Department Seminar, The Wharton School at the University of Pennsylvania
2019	Statistics Department Seminar, University of Southern California, Marshall

2019	Statistics Department Seminar, Rutgers University
2018	Machine Learning Group Seminar, Stanford University
2017	Statistics Department Seminar, HongKong University of Science and Technology
2017	Statistics Department Retreat Seminar, Stanford University
2017	Machine Learning Group Seminar, Stanford University
2016	Statistics Department Retreat Seminar, Stanford University

Graduate Student Advising and Committees

Yunlu Chen	Ph.D. Advisor
Zewei Li	Ph.D. Advisor
Xiaolong Liu	Ph.D. Advisor
Yifei Wang	Master Advisor
Brody Kendall	Ph.D. Committee Member
Tim Tsz-Kit Lau	Ph.D. Committee Member

Undergraduate Student Advising and Committees

Bill Chen	Advisor

Service Commitments at Northwestern University

2025	Department Seminar Organizer
2024	Ph.D. Admission Committee Member
2023	Ph.D. Admission Committee Member
2022	Department Seminar Organizer

Reviewing

Journal Reviewing *Annals of Statistics (AOS)*, *Journal of the American Statistical Association (JASA)*, *Bernoulli*, *Mathematical Programming*, *SIAM Journal of Optimization (SIOPT)*, *Operations Research*, *Mathematics of Operations Research*, *Journal of Machine Learning Research (JMLR)*, *IEEE Transactions on Information Theory (TIT)*, *IEEE Transactions on Signal Processing (TSP)*, *IEEE Journal on Selected Areas in Information Theory (JSAIT)*, *IEEE Transactions on Knowledge and Data Engineering*, *IMA Journal on Numerical Analysis*, *PLOS ONE*, *Frontiers in Psychology*

Conference Reviewing International Conference on Machine Learning (ICML), Neural Information Processing Systems (NIPS), Artificial Intelligence and Statistics (AIStats), Conference on Learning Theory (COLT), ACM Symposium on Theory of Computing (STOC), IEEE International Symposium on Information Theory (ISIT).

Outside Interests

Hiking, piano, card-games, table tennis